

HDOC 23

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Batch: 16

1. Algorithm

1. **Start** main program.
2. Display the menu with options 1 to 5 to the user.
3. Read the user's `choice`.
4. Use a `switch` statement to call the respective function: `solidSquare()`, `solidRectangle()`, `rowNumberPattern()`, `colNumberPattern()`, or exit if choice is 5.
5. **Function** `solidSquare()` :
 - Prompt for and read `n`. Validate `n > 0`.
 - Loop `i` from 1 to `n`. Inside, loop `j` from 1 to `n`.
 - Print `"* "`. After the inner loop, print a newline.
6. **Function** `solidRectangle()` :
 - Prompt for and read `m` and `n`. Validate both `> 0`.
 - Loop `i` from 1 to `m`. Inside, loop `j` from 1 to `n`.
 - Print `"* "`. After the inner loop, print a newline.
7. **Function** `rowNumberPattern()` :
 - Prompt for and read `m` and `n`. Validate both `> 0`.
 - Loop `i` from 1 to `m`. Inside, loop `j` from 1 to `n`.
 - Print `i`. After the inner loop, print a newline.
8. **Function** `colNumberPattern()` :
 - Prompt for and read `m` and `n`. Validate both `> 0`.
 - Loop `i` from 1 to `m`. Inside, loop `j` from 1 to `n`.
 - Print `j`. After the inner loop, print a newline.
9. Repeat steps 2-4 until `choice == 5`.
10. **Stop**

2. Test-Case Table

TC ID	Scenario Description	Inputs	Expected Output
TC-01	Solid square pattern (Valid)	choice: 1, n: 4	4 rows, each with 4 asterisks (* * * *)
TC-02	Solid rectangular pattern (Valid)	choice: 2, m: 3, n: 5	3 rows, each with 5 asterisks (* * * * *)
TC-03	Row-number pattern (Valid)	choice: 3, m: 4, n: 4	Row 1: 1 1 1 1, Row 2: 2 2 2 2, etc.
TC-04	Column-number pattern (Valid)	choice: 4, m: 4, n: 4	Row 1: 1 2 3 4, Row 2: 1 2 3 4, etc.
TC-05	Invalid dimensions (Boundary)	choice: 2, m: -2	Error message: Dimensions must be positive.
TC-06	Invalid menu choice	choice: 9	Invalid choice. Please select 1-5.

3. C Program Source Code

```
#include <stdio.h>

// Function Prototypes (Parameterless, void return type)
void solidSquare(void);
void solidRectangle(void);
void rowNumberPattern(void);
void colNumberPattern(void);

int main() {
    int choice;
    do {
        printf("\n--- Pattern Generation Menu ---\n");
        printf("1. Solid square pattern\n");
        printf("2. Solid rectangular pattern\n");
        printf("3. Row-number pattern\n");
        printf("4. Column-number pattern\n");
        printf("5. Exit\n");
        printf("Enter your choice: ");
        scanf("%d", &choice);

        switch (choice) {
            case 1: solidSquare(); break;
            case 2: solidRectangle(); break;
            case 3: rowNumberPattern(); break;
            case 4: colNumberPattern(); break;
            case 5: printf("Exiting program...\n"); break;
            default: printf("Error: Invalid choice. Please enter 1-5.\n");
        }
    } while (choice != 5);

    return 0;
}

void solidSquare(void) {
    int n, i, j;
    printf("Enter number of rows n: ");
    scanf("%d", &n);
    if (n <= 0) { printf("Error: Dimension must be positive.\n"); return; }

    printf("Solid Square Pattern:\n");
    for (i = 1; i <= n; i++) {
        for (j = 1; j <= n; j++) {
            printf("* ");
        }
        printf("\n");
    }
}

void solidRectangle(void) {
    int m, n, i, j;
    printf("Enter number of rows m and columns n: ");
    scanf("%d %d", &m, &n);
    if (m <= 0 || n <= 0) { printf("Error: Dimensions must be positive.\n"); return; }
```

```

printf("Solid Rectangular Pattern:\n");
for (i = 1; i <= m; i++) {
    for (j = 1; j <= n; j++) {
        printf("* ");
    }
    printf("\n");
}

}

void rowNumberPattern(void) {
    int m, n, i, j;
    printf("Enter number of rows m and columns n: ");
    scanf("%d %d", &m, &n);
    if (m <= 0 || n <= 0) { printf("Error: Dimensions must be positive.\n"); return; }

    printf("Row-number Pattern:\n");
    for (i = 1; i <= m; i++) {
        for (j = 1; j <= n; j++) {
            printf("%d ", i);
        }
        printf("\n");
    }
}

}

void colNumberPattern(void) {
    int m, n, i, j;
    printf("Enter number of rows m and columns n: ");
    scanf("%d %d", &m, &n);
    if (m <= 0 || n <= 0) { printf("Error: Dimensions must be positive.\n"); return; }

    printf("Column-number Pattern:\n");
    for (i = 1; i <= m; i++) {
        for (j = 1; j <= n; j++) {
            printf("%d ", j);
        }
        printf("\n");
    }
}

}

```